EE-133L P ROGRAMMING FUNDAMENTALS

Laboratory Manual

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Registration No. 2025 – EE – 16

Lab Section: (A)

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SUBJECT TITLE:	Programming Fundamentals Lab		

LAB PERFORMANCE EVALUATION RUBRIC

Performance	CLOs Mapping	Excellent (100%–80%)	Good (80%–55%)	Satisfactory (55%–30
Logic / Approach to Problem Solving (30 Points)	**CLO1, CLO2, CLO3.**	Demonstrates clear, innovative, and structured problem-solving approach, selects the most efficient logic and justifies choices effectively.	Applies appropriate logic with minor gaps, solution works with some inefficiency but shows understanding of approach.	Basic attempt to apply log with significant gaps, solu partially correct and lacks
Code Implementation & Error Debugging (40 Points)	**CLO1, CLO2, CLO3.**	Code is well-structured, follows best practices, and compiles successfully on the first attempt; independently debugs errors with precision.	Code is mostly correct with minor issues; requires some debugging but resolves independently.	Code contains multiple er requires assistance of lab instructor for debugging a corrections.
Documentation & Report Writing (20 Points)	**CLO4**	Documentation is comprehensive, professional, and well-formatted, includes clear explanations, comments, and structured reporting.	Documentation is complete with minor formatting or clarity issues; comments are present but limited.	Documentation is minima clarity, or inconsistent formatting; explanations a insufficient.
Discipline & Lab Ethics (10 Points)	**CLO4**	Arrived on time, fully prepared, and used time well. Maintained clean coding environment (file structure, naming, saving work). Followed all lab policies.	Arrived slightly late (≤5 mins), prepared, used time well, and maintained proper coding environment. Generally followed lab rules.	Arrived late (>5 mins but mins), were partially prepa Completed work but show little interest. Needed rem about coding practices/lab discipline.
TOTAL		<------Total points earned (Max 100) = **Lab Performance Grade**		

Experiment No. 2:
Variables and Implementation of Arithmetic Operations in C

Objectives

The objectives of this lab are:

1. To understand the use of variables.
2. To learn about different data types.
3. To learn how to take input from the user.
4. To implement arithmetic operations in C.
5. To understand the significance of in-built libraries; e.g., math.h.

Prerequisites

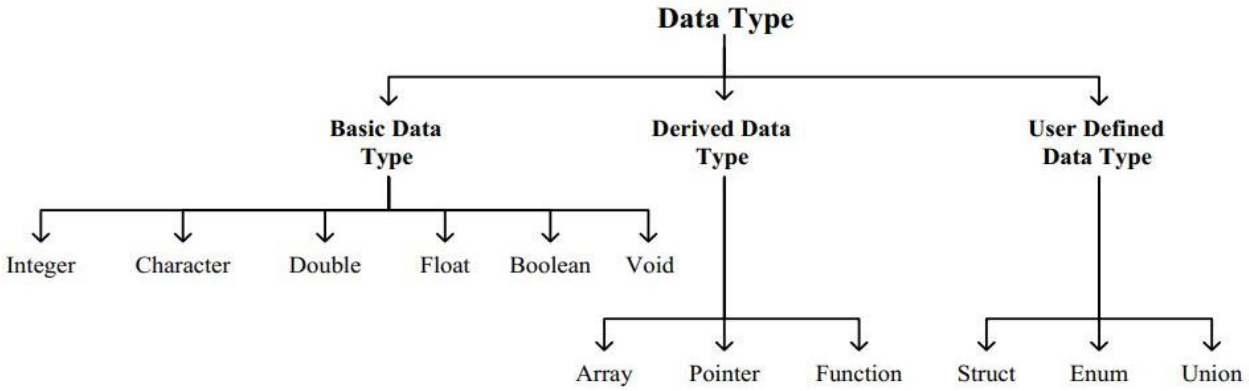
Following are the prerequisites of this lab are:

- Installed Ubuntu.
- Basic knowledge of writing simple program in C language.

Introduction

Variable
A labeled storage in the memory of the computer, which stores a value is called a variable. It is named variable, because it can be updated during the execution of a program.

Data type
The data type of a variable simply describes which kind of data is stored by that variable. Moreover, data type also determines which operation is going to be performed on that data. It is classified as:



The focus of this lab is the utilization of basic data types of variables.

Format Specifier

In programming, each data type requires a unique format specifier to be printed after the execution of the statement. Table 1 presents the list of format specifiers.

Table 1: List of format specifiers

Data Type	Key Words in C	Format Specifier
Integer (signed)	int	%d or %i
Unsigned integer	unsigned int	%u
Character	char	%c
Float	float	%f
Double	double	%lf
Boolean	bool	%d or %i
String (array of character)	char str[ ]	%s

Example

```
#include <stdio.h>
int main() {
    int rollnumber = 20;
    float aggregate = 80.75;
    char section = 'C';
    printf("Age: %d\n", rollnumber);
    printf("GPA: %.2f\n", aggregate);
    printf("Grade: %c\n", section);
    return 0;
}
```

Input-Output Functions

- **scanf Function**
  "scanf( )" function is used to take input from the keyboard in C language programming.
- **printf Function**
  "printf( )" function is used to print the output of the executed statement of the C program.

Example

```
#include <stdio.h>
int main() {
    int age;
    printf("Enter your age: ");
```

```
    scanf("%d", &age);  // '&' address operator
    printf("You age is: %d\n", age);
    return 0;
}
```

Note: "&" is an address operator. When it is combined with the name of variable it provides the address of that stored variable in the memory.

Comparison of %d and %i Format Specifiers in scanf

Code

```
#include <stdio.h>

int main() {
    int value;
    printf("Enter a number: ");
    scanf("%i", &value);  /*change to %d or %i before running to observe the
behavior of %i and %d for scanf*/

    printf("Value stored (decimal): %d\n", value);
    return 0;
}
```

Note: When "%i" is used in aprintf statement, it works exactly like "%d". However, in a scanf statement, "%i" automatically detects the base (number system) of the input.

Output

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab01_2025_EE_016$ ./program
Enter a number: 10
Value stored (decimal): 10
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab01_2025_EE_016$ ./program
Enter a number: 010
Value stored (decimal): 8
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab01_2025_EE_016$ ./program
Enter a number: 0x10
Value stored (decimal): 16
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab01_2025_EE_016$
```

Arithmetic Operations

In the C language, arithmetic operations are performed using arithmetic operators that are mentioned in the following table. Arithmetic operations in C language follow the rule of operator precedence, which are aligned with BODMAS /PEMDAS rules in mathematics.

Table 3: Arithmetic Operations

Operations	Arithmetic Operators	Algebraic Expression	C Expression
Addition	+	$x + y$	x + b
Subtraction	-	$a - b$	a – b
Multiplication	*	cy	c * y
Division	/	$\frac{}{}$	a/b
Remainder	%	$x \bmod y$	x % y

Example

```
#include <stdio.h>
int main() {
    int a;
    int b;
    printf("First integer: ");
    scanf("%d", &a);
    printf("Second integer: ");
    scanf("%d", &b);
    printf("Addition: %d\n", a+b);
    printf("Subtraction: %d\n", a-b);
    printf("Multiplication: %d\n", a*b);
    printf("Division: %d\n", a/b);
    printf("Modulus: %d\n", a%b);

    return 0;
}
```

Introduction to <math.h> Header File

In C language, there is a standard library that provides in-built functions for advanced mathematical calculations. The square root of a number can be directly found by including <math.h> header file and calling the in-built function, as shown in the following code.

Example

```
#include <stdio.h>
#include <math.h>
```

```c
int main() {
    double result = sqrt(144);

    printf("Square root of 144 is: %lf\n", result);
    return 0;
}
```

Lab Tasks

Task 1

Find the area and perimeter of a rectangle. Take the measurements of sides of rectangle from the user.

Code

```c
#include <stdio.h>
int main()
{
double Len, Wid, Area, Per;

printf("Enter the Length of Rectangle: ");
scanf("%lf", &Len);
printf("Enter the Width of Rectangle: ");
scanf("%lf", &Wid);

Area = Len * Wid;
Per = 2*(Len + Wid);

printf("Area is %lf | Perimeter is %lf", Area, Per);
}
```

Output

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ gcc area.c -o task1
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ ./task1
Enter the Length of Rectangle: 2
Enter the Width of Rectangle: 4
Area is 8.000000 | Perimeter is 12.000000muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$
```

Task 2

Fix all the syntax and logical errors in the following code, and write the correct version of this C program with its output.

Bugged Code

```c
#include <stdio.h>
int main() {
    int a b;
    printf("Enter first number: ")

    scanf("%d", a)

    printf("Enter second number: ");
    scanf("%d", b)

    printf("Addition = %d\n", a + b)
    printf("Subtraction = %f\n", a - b);
    printf("Multiplication = %d\n", a * b)
    printf("Division = %d\n", a / b);

    return;
}
```

Corrected Code

```c
#include <stdio.h>


int main(){
int a, b;

printf("Enter A: ");
scanf("%d", &a);

printf("Enter B: ");
scanf("%d", &b);

printf("Addition: %d\n", a + b);
printf("Subtraction: %d\n", a - b);
printf("Multiplication: %d\n", a * b);
printf("Division: %f\n", (float)a / (float)b);

return 0;
}
```

Output

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ gcc operations.c -o task2
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ ./task2
Enter A: 12
Enter B: 14
Addition: 26
Subtraction: -2
Multiplication: 168
Division: 0.857143
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$
```

Task 3

Write a C program to swap values of two variables by using a third variable.

Code

```c
#include <stdio.h>

void main(){

int a = 3, b = 4, temp;

printf("Before Swap\na: %d\nb:%d\n", a, b);
temp = a;
a = b;
b = temp;
printf("After Swap\na: %d\nb:%d\n", a, b);

}
```

Output

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ gcc swap.c -o task3
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ ./task3
Before Swap
a: 3
b:4
After Swap
a: 4
b:3
```

Task 4

Write a C program to find the distance between two pints by using in-built functions of math.h library.
Hint: Lets P1= (x1, y1) and P2= (x2, y2). Then, find D = $\sqrt{(2-1)^2+(2-1)^2}$. You can use in-built functions for square root and power.

Code

```
#include <stdio.h>
#include <math.h>
int main(){

double x1 = 1, x2 = 2, y1 = 1 , y2 = 2;
double dist;
dist = sqrt((pow(x2 - x1, 2)) + ((pow(y2 - y1, 2))));
printf("The Distance is: %lf", dist);
}
```

Output

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ gcc distance.c -o dist -lm
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$ ./dist
The Distance is: 1.414214muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab02_2025_EE_016$
```

Extension

Bonus Task 1

Write a program to evaluate the following expression when a=10, b=5, c=15, d=20 and = 45°. The program should work correctly for any values of the variables Don't forget to include the header <math.h> to use trigonometric and pow() functions. Use 22/7 as value of .The output should be as shown below.

$$\frac{\sqrt{\left(\frac{a}{b}sin\theta\right)^{\left(\frac{a}{b^2}\right)} + \left(\frac{b}{a}cos\theta\right)^{\left(\frac{1}{a+b}\right)}}}{(c * tan\theta) - \left(\frac{d}{c}\right)}$$

Bonus Task 2

Write a C program to calculate power of an number without using pow() functions.

Bonus 1:

```c
#include <stdio.h>
#include <math.h>
#define PI ((double)22/(double)7)
int main(){
double a = 10, b = 5, c = 15, d = 20, angle = 45*(PI / 180);
double num, denom;
num = sqrt((pow((a/b)*(sin(angle)),a/(b*b))) + (pow((b/a)*(cos(angle)),1/(a+b))));
denom = (c * tan(angle)) - (d/c);

printf("Computed value = %lf", (num / denom));
}

```

Bonus 2:

```c
#include <stdio.h>

int main(){
int num, power, result = 1;
printf("Enter number: ");
scanf("%d", &num);
printf("Enter power ");
scanf("%d", &power);

for(int i = 1; i <= power; i++){
result *= num;
}
printf("Answer of %d to the power %d is %d", num, power, result);
}
```